

# RICHARD NIRAGIRE RUTAGENGWA

## Digital Health & Health Information Systems Technical Advisor

DHIS2 | Public-Health Surveillance | Immunisation | EMR | Data for Action

Kigali, Rwanda | +250 788 426 156 | rrutagengwa@gmail.com | International & remote assignments

### EXECUTIVE PROFILE

Digital health and health information systems professional with 18 years of progressive experience across public-health data management, DHIS2 design and implementation, electronic medical records, surveillance, immunisation, data triangulation, analytics, and capacity development. Currently Technical Advisor for Information Systems Management at HISP Rwanda, supporting government and partner programmes in Rwanda, the Republic of the Congo, Comoros, and Chad. Combines field-level service experience with national and multi-country delivery, translating programme requirements into practical digital workflows, dashboards, and sustainable user practices.

**NATIONAL-SCALE IMPLEMENTATION • MULTI-COUNTRY DELIVERY • GOVERNMENT COORDINATION • CAPACITY DEVELOPMENT**

### PROFESSIONAL EXPERIENCE

**HISP Rwanda** | Kigali, Rwanda

**Dec 2024 - Present**

#### Technical Advisor - Information Systems Management

Design and implement DHIS2 Tracker and Aggregate solutions for surveillance, immunisation, education, climate-health, and programme management, including metadata, program rules, indicators, dashboards, validation, notifications, and user workflows.

Lead requirements analysis, configuration, testing, deployment readiness, technical documentation, validation, and post-deployment support in collaboration with ministries, RBC, UNICEF, WHO, and implementation partners.

Deliver national and multi-country demonstrations, Training of Trainers, cascade training, coaching, and user support for central, regional, district, hospital, and frontline teams.

#### SELECTED DIGITAL HEALTH PORTFOLIO

**Rwanda | EMS, IDSR & Impuruza** - Designed and supported implementation of DHIS2-based national event and case-based surveillance workflows, risk assessment, notifications, reporting, and rollout; trained IDSR focal persons, PHEOC teams, hospital directors, and surveillance users across the country.

**Rwanda | VPD Data Triangulation** - Triangulated data from multiple DHIS2 sources and supported subdistrict immunity-gap and programme-performance dashboards, data-quality review, field mentorship, and data-to-action discussions with RBC and MoH leadership.

**Comoros | MEV Digitalisation** - Supported end-to-end digitalisation of vaccine-preventable disease surveillance from notification through investigation, laboratory results, clinical follow-up, and final classification, including indicators, dashboards, testing, deployment, and user support.

**Comoros | REV Extension** - Contributed to metadata harmonisation, DHIS2 integration, vaccination reminders, digital supervision tools, rollout, capacity development, and dashboard updates for the Electronic Vaccination Register.

**Rwanda | eBuzima EMR** - Supported Ministry of Health implementation of the Frappe/ERPNext-based EMR through clinical-workflow assessment, staff training, site-readiness review, testing, and go-live support at CorUnum Health Centre, with preliminary rollout support at Rugarama Health Centre.

**Rwanda & Chad | Education and Other Systems** - Contributed to SEMIS/Karame, EPI microplanning, climate-health and malaria early-warning work, portfolio management, REST API and JSON testing, metadata imports, analytics verification, user guidance, and production readiness.

### EARLIER PROFESSIONAL EXPERIENCE

**HISP Rwanda** | Kigali, Rwanda

**Feb 2024 - Dec 2024**

#### Software Developer

Supported eBuzima implementation through workflow assessment, staff training, site readiness, testing, go-live support, and preparation of training materials and a practical demonstration environment.

Built practical knowledge of Frappe/ERPNext architecture, data flows, customisation pathways, and integration opportunities for digital-health solutions.

**Rugarama Health Centre** | Kigali, Rwanda

Aug 2012 - Feb 2024

**Data Manager / Health Information Systems & ICT Lead**

Managed routine health information across clinical services and produced daily, weekly, monthly, quarterly, and annual reports for district, hospital, and national levels, ensuring completeness, timeliness, consistency, and accuracy.

Administered and supported Rwanda HMIS/DHIS2, OpenMRS, e-TB, CRVS, eLMIS, SISCOM, FBF, eTracker, IDSR, MediSOFT, databases, servers, networks, backups, secure access, and end-user support.

Led data validation and Data Quality Assessments, reconciled source documents and patient records, supported external reviews, and worked with providers to improve data integrity and use.

Designed and developed AKABANDO, a PHP/MySQL/Bootstrap solution connecting laboratory, outpatient, pharmacy, and related workflows to reduce reliance on paper registers and strengthen digital records.

**Nyarugenge District Hospital - COVID-19 Treatment Centre** | Kigali, Rwanda

Oct 2020 - Mar 2021

**Data Manager - Temporary Assignment**

Managed treatment-centre data, verification, reporting, and information flow to support timely operational and public-health decision-making.

**Earlier ICT Roles** | Rwanda

2008 - 2012

**ICT Department Lead / IT Technician - EL-TECH and Last Computer Company**

Supervised workshop and field assignments and delivered installation, configuration, maintenance, and repair of computers, servers, networks, Cisco devices, LAN infrastructure, PABX systems, printers, and related ICT equipment.

**CAPACITY BUILDING, SPEAKING & PROFESSIONAL ENGAGEMENT**

Designed and delivered national and multi-country demonstrations, ToT, cascade training, coaching, and post-training support across Rwanda, the Republic of the Congo, and Comoros for government, district, hospital, and frontline users.

Co-facilitated a UNICEF-supported 10-day DHIS2 and Advanced Excel programme in Brazzaville for PEV data managers representing all 52 health districts, covering data entry and validation, visualisation, maps, dashboards, data quality, and data-to-action.

Speaker and DHIS2 technical demonstrator at the Science of Defeating Malaria leadership course in Kigali, presenting Rwanda's community-surveillance and digital malaria response work to an international public-health audience.

Participated in the HISP Rwanda DHIS2 Academy, completed a portfolio of DHIS2 online courses, and contributed to climate-health modelling, predictive analytics, malaria, surveillance, and early-warning knowledge exchanges.

**EDUCATION & CREDENTIALS**

**Master of Science in Software Engineering** | Mount Kenya University - In progress

**Bachelor of Science in Computer Science** | Kigali Independent University (ULK), 2016

**Technical Qualification in Electronics and Telecommunications** | ADB Nyarutarama, 2004

**DHIS2 Credentials** | DHIS2 Academy certificate and portfolio of DHIS2 online-course certificates; supporting certificates available.

**Selected Public-Health Training** | Rwanda HMIS/DHIS2, active case-based surveillance, IDSR, OpenMRS, CRVS, eLMIS, PEPFAR/CDC MER indicators, DQA, Quality Improvement and Assurance, TB, RBF, and maternal and child health.

**TECHNICAL COMPETENCIES**

**DHIS2 & Digital Health** | Tracker and Aggregate configuration; metadata; program rules and variables; indicators; dashboards; maps; validation; notifications; testing; deployment; REST API testing; JSON import/export; eBuzima/Frappe/ERPNext; OpenMRS.

**Data & Implementation** | Data triangulation; DQA; analysis and visualisation; advanced Excel; requirements analysis; workflow design; stakeholder coordination; training facilitation; issue tracking; production readiness; technical writing; multi-country collaboration.

**Development & Systems** | PHP; MySQL; Bootstrap; JavaScript; database administration; backups; Linux fundamentals; Windows administration; TCP/IP; LAN/WAN; wireless networks; routers and switches; VPN; Active Directory; servers; fibre optics; end-user support.

**LANGUAGES & MOBILITY**

**Kinyarwanda** - Native | **English** - Professional proficiency | **French** - Professional proficiency

**Country experience:** Rwanda, Republic of the Congo, Comoros, and Chad. Available for remote work, international assignments, and periodic travel. References available upon request.